

# Ema Maria Monteiro Martins

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Master's student in Informatics and Computing Engineering with experience across **backend**, **frontend**, and **mobile** development. Skilled in **C#**, **Python**, and **JavaScript**, with hands-on experience building **full-stack applications**, **real-time systems**, and **AI-integrated software**.

## Education

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|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Universidade do Porto<br>Master's Degree in Informatics and Computing Engineering                                          | Sep 2024 - Jul 2026<br>Porto, Portugal |
| Universidade do Minho<br>Bachelor's Degree in Informatics Engineering (completed concurrently with Biomedical Engineering) | Sep 2022 - Jul 2024<br>Braga, Portugal |
| Universidade do Minho<br>Bachelor's Degree in Biomedical Engineering (completed concurrently with Informatics Engineering) | Sep 2020 - Jul 2024<br>Braga, Portugal |

## Experience

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| Mindera<br>Master Thesis Intern                                                                                                                                                                                                                                                                                                                                                                                    | Feb 2026 - Jun 2026<br>Porto, Portugal |
| <ul style="list-style-type: none"><li>Developed an internal project engineering an <b>AI agent</b> to act as a believable second player in two-player video games, prioritizing <b>human-like behavior</b> over flawless performance.</li><li>Integrated computer vision and AI models (<b>NVIDIA NitroGen</b>, <b>YOLO</b>, <b>Grounding DINO</b>) to process in-game environments in <b>real-time</b>.</li></ul> |                                        |

## Projects

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| Finalyze   React Native, Spring Boot, PostgreSQL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repo Link |
| <ul style="list-style-type: none"><li>Collaborated on the strategic and technical development of <b>Finalyze</b>, a <b>fintech mobile app</b> designed to help users track expenses, set savings goals and build financial habits through <b>gamification</b> and <b>data-driven benchmarking</b>.</li><li>Co-authored key strategic documents — <b>Business Model Canvas (BMC)</b>, <b>Value Proposition Canvas (VPC)</b>, <b>Product Vision Board</b>, <b>Project Charter</b> and <b>Strategic Plan</b>, following <b>agile project management</b> practices.</li></ul> |           |
| Car Mechanic Interface   Vue.js, JSON Server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Repo Link |
| <ul style="list-style-type: none"><li>Designed and developed an <b>auto repair shop interface</b> tailored for mechanics to streamline daily service workflows.</li><li>Implemented core functionality to execute, pause, and track repair services, including logging detailed explanations for task blockers.</li><li>Integrated user management features enabling mechanics to access and update personal profile information directly within the system.</li></ul>                                                                                                    |           |
| Receitas   Kotlin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Repo Link |
| <ul style="list-style-type: none"><li>Developed a <b>mobile app</b> ("Receitas") to digitize personal recipe management and replace paper-based tracking.</li><li>Implemented full CRUD operations allowing users to efficiently add, view, update, and delete custom recipes.</li><li>Engineered a <b>JSON</b> export feature to support data portability, backups, and seamless integration with other applications.</li></ul>                                                                                                                                          |           |
| Clothing Recommendation System   Python Flask                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repo Link |
| <ul style="list-style-type: none"><li>Developed a <b>Clothing Recommendation System</b> using a custom <b>KD-Tree algorithm</b> and <b>Euclidean weighting</b>, matching suggestion quality of major e-commerce platforms such as Primark and Shein.</li><li>Built a full-stack <b>Flask web application</b>, optimizing high-dimensional <b>one-hot encoded data</b> for fast, transparent product retrieval.</li></ul>                                                                                                                                                  |           |

## Skills

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|------------------------------------------------------------------------|--------------------|
| Technical Skills                                                       | Languages          |
| Backend: C, C#, C++, Python, Java, Haskell, .NET                       | Portuguese: Native |
| Frontend: JavaScript, React, Vue.js, HTML/CSS, Pug                     | English: Fluent    |
| Mobile: Kotlin, React Native, Flutter                                  | Spanish: Basic     |
| Databases: MongoDB, MySQL, Redis, PostgreSQL, Neo4j                    |                    |
| Practices: Git, Agile/Scrum                                            |                    |
| Game Dev: Godot, Unity                                                 |                    |
| AI & Machine Learning: Unity ML-Agents, NitroGen, YOLO, Grounding DINO |                    |

## Leadership & Volunteering

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|---------------------------------------------------------------------------------------------------------|---------------------|
| Vice-president of Literatuna – University of Minho                                                      | Sep 2023 – Jul 2024 |
| Student Organization Leadership: Coordinated activities and performances of a university musical group. | Braga, Portugal     |
| Erasmus+ Project – ANIMA 2022                                                                           | May 2022            |
| International Sociocultural Event: International meeting of sociocultural animation.                    | Portugal / Europe   |
| Volunteer – Extracurricular Classes                                                                     | Sep 2022 – Mar 2023 |
| Volunteering: Taught extracurricular classes to children aged 6–9, promoting environmental awareness.   | Braga, Portugal     |